LAB 10

Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	Strategy search
TSC reference	A2, A4, K2
Duration	20 min
Folder	labs_fullcode/Lab10-strategy-grid-search/
Files	Lab10-strategy-grid-search/search_stage.py, pipeline/search.py, Lab10-strategy-grid-search/solution/search.py
Prompts	Lab10_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

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Scenario

A junior analyst wants the pipeline to 'find the best parameters'. The head of research agrees on one condition: parameters are ranked on 2018-2021 only and judged on the untouched 2022-2025 period (including the 2022 bear market), and the stage must report how much of the in-sample result decays.

Objective

Build `python -m pipeline search` to grid-search three signal families, rank parameter sets in sample, measure out-of-sample decay, prefer a robust parameter region, and decide whether anything deserves promotion into config.yaml.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 10	python3 Lab10-strategy-grid-search/search_stage.py (same as: python -m pipeline search)
The complete code	pipeline/search.py (reference copy: Lab10-strategy-grid-search/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab10-strategy-grid-search/Lab10_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: Define the search space (A2, K2)

Read GRIDS in Lab10-strategy-grid-search/solution/search.py: sma_cross fast x slow (24 valid pairs), rsi_reversion n x entry x exit (36), bollinger_breakout window x k (16) - 76 sets.

Check: You can say why fast 50 / slow 50 is skipped and why every extra set increases the chance of a lucky winner.

Step 2: Build the grid search (A2, A4)

Ask your assistant for the search engine.

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/search.py. IS_END = '2021-12-31', OOS_START = '2022-01-01'. GRIDS = {'sma_cross': {'fast': [5,10,20,30,50], 'slow': [50,100,150,200,250]}, 'rsi_reversion': {'n': [2,3,5,14], 'entry': [10,20,30], 'exit': [50,60,70]}, 'bollinger_breakout': {'window': [10,20,30,50], 'k': [1.0,1.5,2.0,2.5]}}. grid_points(name) yields parameter dicts (skip sma_cross with fast >= slow). search(bars, costs) runs ONE backtest per set with pipeline.signals.target_position and common.backtest.backtest (signals are causal, so one run covers both windows), then records IS and OOS Sharpe and CAGR (common.metrics) and trades; add IS_rank and OOS_rank (method='min', NaN last) and decay = OOS - IS Sharpe. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: search(...) returns a DataFrame with 76 rows.

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Step 3: Add ranking, robustness and the decay report (A4, K2)

Ask your assistant for run() and main().

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/search.py add robust_choice(ma) - for the sma_cross grid, the cell with the best AVERAGE in-sample Sharpe over itself and its neighbours - and run(symbol='SPY') for `python -m pipeline search`: print buy-and-hold Sharpe IS and OOS; a decay report with the best IS set per family (IS -> OOS Sharpe, decay, OOS rank of 76, trades); the top 5 by IS Sharpe; the Spearman rank correlation between IS and OOS Sharpe; the average decay of the top 5; the share of all sets that beat buy and hold OOS; the single best MA cell versus the robust MA region; and a suggested config.yaml strategy line (do NOT edit the config automatically). Save _outputs/Lab10_grid_results.csv, _outputs/Lab10_decay_report.csv and chart Lab10_grid_heatmap.png (IS and OOS Sharpe heatmaps of the MA grid with values in each cell). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline search --help shows --symbol.

Step 4: Run the search (A4)

Run: python -m pipeline search --symbol SPY (same as: python3 Lab10-strategy-grid-search/search_stage.py).

Check: 76 sets; buy-and-hold Sharpe 0.87 IS and 0.67 OOS. Best per family: sma_cross 10/100 IS 1.23 -> OOS 0.70 (decay -0.53, OOS rank 26); rsi_reversion n 2, entry 30, exit 70: 0.60 -> 0.74; bollinger_breakout 20 / 1.0: 0.91 -> 1.04 (OOS rank 3).

Step 5: Read the overfitting evidence (A4, K2)

Read the top-5 table and the three summary lines.

Check: Top 5 are all sma_cross; average decay -0.62 Sharpe points; IS/OOS rank correlation 0.06 (the in-sample ranking has almost no predictive value); 43% of all sets beat buy and hold out of sample.

Step 6: Prefer a robust region to a single peak (A2, A4)

Compare the single best MA cell with the robust MA region and open _outputs/Lab10_grid_heatmap.png.

Check: Best cell 10/100: IS 1.23 -> OOS 0.70; robust region 5/50: neighbour-average IS 1.04, own IS 0.92 -> OOS 0.44 - robustness reduces luck but guarantees nothing.

Step 7: Decide whether to promote anything (A2, K2)

Do not edit config.yaml yet. Weigh the evidence first.

AI prompt - paste into your AI coding assistant
Using labs_fullcode/_outputs/Lab10_decay_report.csv and labs_fullcode/_outputs/Lab10_grid_results.csv, argue for and against replacing the pipeline's configured ema_cross 10/30 with the best sma_cross or bollinger_breakout set. Mention the number of sets tried (76), the decay, the rank correlation and what the diagnostics stage (Lab 18) should check before any change. Finish with a one-line recommendation.

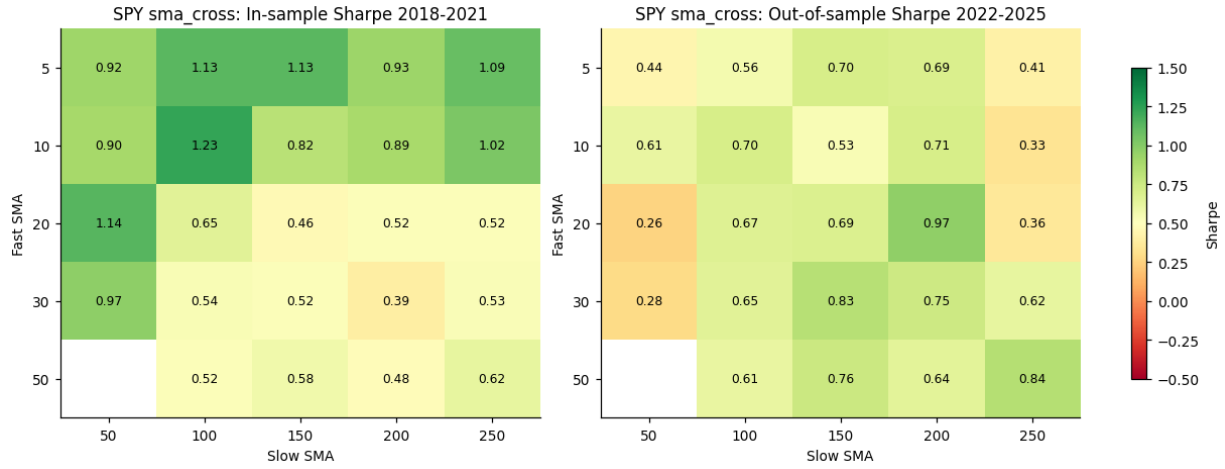
Check: A balanced recommendation that keeps ema_cross until a diagnostics run and an A/B test (Lab 13) support a change.

Expected Output (Full Code)

python -m pipeline search --symbol SPY: 76 parameter sets; buy-and-hold Sharpe 0.87 in sample and 0.67 out of sample; best sma_cross (10/100) 1.23 -> 0.70; best rsi_reversion (2, 30, 70) 0.60 -> 0.74; best bollinger_breakout (20, 1.0) 0.91 -> 1.04; top-5 average decay -0.62; IS/OOS rank correlation 0.06;

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43% of sets beat buy and hold out of sample; files _outputs/Lab10_grid_results.csv, _outputs/Lab10_decay_report.csv and chart _outputs/Lab10_grid_heatmap.png.



Lab 10 output (labs_fullcode/_outputs/Lab10_grid_heatmap.png), offline mode on the bundled data.

Debrief

Grid search is a statistical search over model structures (A2, A4); comparing in-sample with out-of-sample results (K2) exposes how much of the 'best' result was luck. A hold-out period, a count of trials and a preference for robust regions are the pipeline's defences against fooling itself.

Troubleshooting

Symptom	What to do
IntCastingNaNError on ranks	Some sets never trade in one window (NaN Sharpe); rank with na_option='bottom'.
Heatmap has a blank cell	Expected: fast 50 / slow 50 is not a valid pair.
Different results online	Alpaca's IEX data differ slightly from the bundled data; compare patterns, not decimals.

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